Keldysh and Mori–Zwanzig

Two languages for the same open-system dynamics:
a working note and a self-study plan

Notes prepared with Claude

30 September 2026

About the references. All bibliography entries were checked on 30 September 2026 against publisher, Crossref, or arXiv records, and DOIs were added where they exist. The equations in the main text were derived and cross-checked independently of the references.

The thesis in three lines

1. *Density-matrix level.* Ordering Liouville-space superoperators along a single real time axis *is* Keldysh contour ordering. With the Nakajima projector $\mathcal{P} = \rho_B \otimes \mathrm{Tr}_B$ and a Gaussian bath, the NZ memory kernel is the *irreducible self-energy* of the reduced propagator, and the NZ equation is a Dyson equation.
2. *Operator level.* Mori's projection onto a system operator A yields a generalized Langevin equation. For $A = c$ with the anticommutator inner product, the Mori memory function is (up to a factor $-i$) the *retarded* self-energy Σ^R.
3. *Common ground.* Both expansions are built from the two bath kernels that appear in the Keldysh/Feynman–Vernon influence functional: the response χ (retarded, c–q) and the noise S (Keldysh, c–c). Differences between the methods come from *which object is projected* and *how the series is resummed*, not from different physics.

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1 Setup and conventions

We set $\hbar = 1$ throughout and take

$$H = H_S + H_B + H_I, \qquad H_I = \sum_\alpha S_\alpha \otimes B_\alpha, \tag{1}$$

with Hermitian S_α, B_α, a stationary bath state ρ_B ($[H_B, \rho_B] = 0$), and $\langle B_\alpha \rangle \equiv \mathrm{Tr}_B(B_\alpha \rho_B) = 0$. The Liouvillian is $\mathcal{L}X = -i[H, X] = (\mathcal{L}_S + \mathcal{L}_B + \mathcal{L}_I)X$. For a factorized initial state $\rho(0) = \rho_S(0) \otimes \rho_B$, the reduced dynamics defines the dynamical map

$$\rho_S(t) = \Pi(t)\, \rho_S(0), \qquad \Pi(t)X \equiv \mathrm{Tr}_B\big[\mathrm{e}^{\mathcal{L}t}(X \otimes \rho_B)\big]. \tag{2}$$

Left/right superoperators and the Keldysh rotation. For any operator A, define $A^+X = AX$ (acting on the ket, i.e. the *forward* branch) and $A^-X = XA$ (acting on the bra, i.e. the *backward* branch). Then $\mathcal{L}_I = -i(H_I^+ - H_I^-)$. The Keldysh rotation to "classical" and "quantum" combinations is

$$A^{\mathrm{c}} = \tfrac{1}{2}\big(A^+ + A^-\big) \ \ \big(A^{\mathrm{c}}X = \tfrac{1}{2}\{A, X\}\big), \qquad A^{\mathrm{q}} = A^+ - A^- \ \ \big(A^{\mathrm{q}}X = [A, X]\big). \tag{3}$$

Since system and bath superoperators commute, $S^+B^+ - S^-B^- = S^{\mathrm{c}}B^{\mathrm{q}} + S^{\mathrm{q}}B^{\mathrm{c}}$, and therefore

$$\boxed{\mathcal{L}_I = -i \sum_\alpha \big(S^{\mathrm{q}}_\alpha B^{\mathrm{c}}_\alpha + S^{\mathrm{c}}_\alpha B^{\mathrm{q}}_\alpha\big).} \tag{4}$$

Two trace identities do most of the work: $\mathrm{Tr}_B(B^{\mathrm{q}}Y) = 0$ for any Y (the trace of a commutator vanishes), and $\mathrm{Tr}_B(B^{\mathrm{c}}Y) = \mathrm{Tr}_B(BY)$.

Bath kernels. With $B_\alpha(t) = \mathrm{e}^{iH_Bt} B_\alpha \mathrm{e}^{-iH_Bt}$,

$$S_{\alpha\beta}(t) = \tfrac{1}{2}\langle\{B_\alpha(t), B_\beta(0)\}\rangle \qquad \text{(symmetrized noise; Keldysh component)}, \tag{5}$$

$$\chi_{\alpha\beta}(t) = i\,\theta(t)\langle[B_\alpha(t), B_\beta(0)]\rangle \qquad \text{(Kubo response; retarded component)}. \tag{6}$$

The full correlator is $C_{\alpha\beta}(t) = \langle B_\alpha(t) B_\beta\rangle = S_{\alpha\beta}(t) - \frac{i}{2}\chi_{\alpha\beta}(t)$ for $t > 0$. For a single Hermitian B, $S(t)$ is real and even. In thermal equilibrium the fluctuation–dissipation theorem (FDT) reads $S(\omega) = \coth(\beta\omega/2)\,\chi''(\omega)$, with $S(\omega) = \int \mathrm{d}t\, \mathrm{e}^{i\omega t} S(t)$ and $\chi'' = \mathrm{Im}\,\chi$. Up to factors of i, these are the bath Green's functions D^K and D^R of Keldysh field theory [1, 2].

2 Keldysh on the density matrix: why the contour is already there

In the interaction picture, $\tilde{\rho}(t) = T \exp\big[\int_0^t \mathrm{d}s\, \tilde{\mathcal{L}}_I(s)\big]\rho(0)$. Expanding the time-ordered exponential produces strings of superoperators $H_I^{\pm}(t_n) \cdots H_I^{\pm}(t_1)$ with $t_n > \cdots > t_1$. A "+" factor sits to the left of $\rho(0)$ in time order, and a "−" factor sits to the right in *anti*-time order. This is precisely the Keldysh contour of Fig. 1 [3, 4]. Keldysh ordering is therefore nothing more than ordinary time ordering in Liouville space.

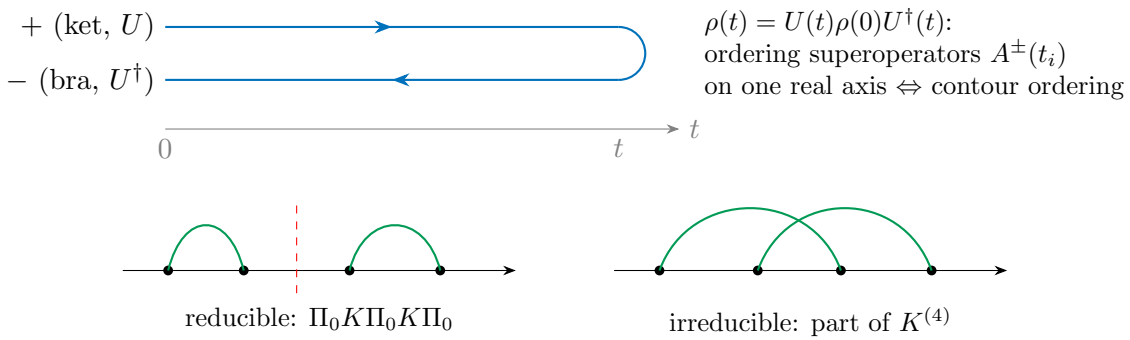


Figure 1: Top: the Keldysh contour. The forward and backward branches are the ket and bra sides of ρ. Bottom: real-time diagrams for the reduced propagator. Vertices are system superoperators $S^{\mathrm{c/q}}$ and lines are bath contractions (S or χ). A diagram that can be cut between vertices without cutting a line (red) is reducible. The NZ kernel collects exactly the irreducible diagrams.

Wick's theorem for superoperators. For a Gaussian bath, $\mathrm{Tr}_B[\tilde{B}^{a_n}(t_n)\cdots\tilde{B}^{a_1}(t_1)\rho_B]$ (with $a_i \in \{\mathrm{c},\mathrm{q}\}$) factorizes into pairwise contractions. For $t > s$ these are

$$\mathrm{Tr}_B[B^{\mathrm{c}}(t)B^{\mathrm{c}}(s)\rho_B] = S(t-s), \qquad \mathrm{Tr}_B[B^{\mathrm{c}}(t)B^{\mathrm{q}}(s)\rho_B] = -i\chi(t-s), \qquad \mathrm{Tr}_B[B^{\mathrm{q}}(t)\,\cdot\,] = 0. \tag{7}$$

This is the familiar Keldysh causality structure: $\langle\phi_{\mathrm{c}}\phi_{\mathrm{c}}\rangle \propto D^K$, $\langle\phi_{\mathrm{c}}\phi_{\mathrm{q}}\rangle \propto D^R$, $\langle\phi_{\mathrm{q}}\phi_{\mathrm{q}}\rangle = 0$ [1]. By Eq. (4), a B^{c} vertex carries S^{q} and a B^{q} vertex carries S^{c}. Hence

System-side Feynman rules

Noise lines $S(t-s)$ connect $S^{\mathrm{q}}(t)$ to $S^{\mathrm{q}}(s)$. Response lines $-i\chi(t-s)$ connect the *later* $S^{\mathrm{q}}(t)$ to the *earlier* $S^{\mathrm{c}}(s)$. There are no lines of c–c type on the system side, the analogue of the vanishing qq Keldysh propagator. The latest vertex is always a commutator S^{q}, which preserves Tr_S (trace preservation of Π).

Summed over all pairings, the same rules generate the Feynman–Vernon influence functional [5–7]. In Keldysh-rotated variables,

$$\Phi[s^{\mathrm{c}}, s^{\mathrm{q}}] = -\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \left[s^{\mathrm{q}}(t_1) S(t_1-t_2) s^{\mathrm{q}}(t_2) - i\, s^{\mathrm{q}}(t_1)\chi(t_1-t_2)s^{\mathrm{c}}(t_2)\right], \tag{8}$$

where $s^\pm$ are the path variables on the two branches. S is the Feynman–Vernon noise kernel and χ the dissipation kernel. For a single Gaussian bath this equation contains *all* the information that any of the methods below will use.

3 Nakajima–Zwanzig and TCL as Dyson equations

3.1 The NZ equation

Take $\mathcal{P}X = \rho_B \otimes \mathrm{Tr}_B\, X$ and $\mathcal{Q} = 1 - \mathcal{P}$ [8–10]. Our assumptions give $\mathcal{P}\mathcal{L}_I\mathcal{P} = 0$ and $[\mathcal{P}, \mathcal{L}_S + \mathcal{L}_B] = 0$. For a factorized initial state the inhomogeneous term vanishes, and

$$\dot{\rho}_S(t) = \mathcal{L}_S\rho_S(t) + \int_0^t \mathrm{d}s\, K(t-s)\,\rho_S(s), \qquad K(\tau)X = \mathrm{Tr}_B\left[\mathcal{L}_I\, \mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\, \mathcal{Q}\mathcal{L}_I\,(X\otimes\rho_B)\right]. \tag{9}$$

In Laplace space, $\hat{f}(z) = \int_0^\infty \mathrm{d}t\, \mathrm{e}^{-zt} f(t)$ and

$$\boxed{\hat{\Pi}(z) = \frac{1}{z - \mathcal{L}_S - \hat{K}(z)} \quad \Longleftrightarrow \quad \hat{\Pi} = \hat{\Pi}_0 + \hat{\Pi}_0\, \hat{K}\, \hat{\Pi}, \qquad \hat{\Pi}_0 = (z - \mathcal{L}_S)^{-1}.} \tag{10}$$

3.2 Why K is the irreducible self-energy

Insert the Wick expansion of Section 2 into $\Pi(t)$. Every contribution is a time-ordered string of system superoperators, dressed by contraction lines and interleaved with $\mathrm{e}^{\mathcal{L}_S \Delta t}$. Now cut the time axis between any two consecutive vertices. If no line crosses the cut, the bath returns to ρ_B there and the diagram factorizes as $\cdots \Pi_0\, \mathcal{P} \cdots$. The factor $\mathcal{Q}$ inside $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}$ in Eq. (9) removes exactly these returns to ρ_B. Hence

Identity

$K = \sum\{\text{irreducible real-time diagrams}\}$, and Eq. (10) is the Dyson equation of Keldysh real-time diagrammatics for the reduced density matrix [11–15].

For fermionic reservoirs the same statement holds, with sign factors from line crossings; this is the Schoeller–Schön setting. For non-Gaussian baths, higher cumulants replace pairwise lines. In both cases irreducibility is defined with respect to the *reference state* ρ_B that appears in $\mathcal{P}$. Changing the projector (for example, to correlated projectors) changes the kernel.

3.3 Second order: the Keldysh structure made explicit

At order $\mathcal{L}_I^2$ we can drop $\mathcal{Q}$ in Eq. (9) because $\mathcal{P}\mathcal{L}_I\mathcal{P} = 0$. Using Eqs. (4) and (7),

$$\boxed{K^{(2)}(\tau) = -\sum_{\alpha\beta} S_\alpha^{\mathrm{q}}\, \mathrm{e}^{\mathcal{L}_S \tau}\Big[S_{\alpha\beta}(\tau)\, S_\beta^{\mathrm{q}} - i\, \chi_{\alpha\beta}(\tau)\, S_\beta^{\mathrm{c}}\Big].} \tag{11}$$

Written out, this is the familiar Born kernel

$$K^{(2)}(\tau)X = -\sum_{\alpha\beta}\big[S_\alpha, \mathrm{e}^{\mathcal{L}_S\tau}\big(C_{\alpha\beta}(\tau)S_\beta X - C_{\beta\alpha}(-\tau)XS_\beta\big)\big]. \tag{12}$$

The rotated form in Eq. (11) shows the two separate bath inputs, noise and response, and so connects directly to the influence functional (8).

Markov and Lindblad limits. Replace $\rho_S(s) \to \mathrm{e}^{-\mathcal{L}_S(t-s)}\rho_S(t)$ and extend the upper limit of integration to ∞; this gives Redfield's equation. The secular approximation then gives the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) form, with rates set by $C(\omega) = S(\omega) + \chi''(\omega)$, which equals $2\chi''(\omega)[1 + n_B(\omega)]$ in equilibrium. In the field-theory language, a Lindblad equation corresponds to a Keldysh action with time-local Σ^R and frequency-independent Σ^K [16]:

$$S_K = \int \mathrm{d}t\Big[\bar{\psi}_+ i\partial_t\psi_+ - \bar{\psi}_- i\partial_t\psi_- - H_+ + H_- - i\sum_j\Big(L_{j+}\bar{L}_{j-} - \tfrac{1}{2}\big(\bar{L}_{j+}L_{j+} + \bar{L}_{j-}L_{j-}\big)\Big)\Big]. \tag{13}$$

3.4 TCL and the NZ–TCL map

The time-convolutionless (TCL) generator [17, 18] is $\mathcal{G}(t) = \dot{\Pi}(t)\Pi(t)^{-1}$, defined wherever $\Pi(t)$ is invertible. Substituting Eq. (9) gives

$$\mathcal{G}(t) = \mathcal{L}_S + \int_0^t \mathrm{d}s\, K(t-s)\,\Pi(s)\,\Pi(t)^{-1}, \tag{14}$$

an exact relation. Nestmann, Bruch and Wegewijs turn it into a fixed-point equation that generates $\mathcal{G}$ from K (and conversely) [19]. Reimer *et al.* compare five exact formulations, with attention to complete positivity and divisibility [20]. Three facts are worth keeping in mind:

- Truncating K and truncating $\mathcal{G}$ at the same order are *different* approximations. They agree at second order and differ from fourth order on [14, 15, 21].
- $\mathcal{G}(t)$ can be singular when $\Pi(t)$ is not invertible (for example, the resonant damped Jaynes–Cummings model at strong coupling [10]); K never is. "Local versus nonlocal" is therefore a question about the *representation*, not about the physics [22].
- With correlated initial states, the NZ inhomogeneity reappears. On the Keldysh side it corresponds to the imaginary-time (Matsubara) leg of the Kadanoff–Baym contour [23].

4 Mori's projection and the retarded self-energy

4.1 Generalized Langevin equation

Mori projects *operators*, not states [24–26]. Let $\dot{A} = i[H, A] \equiv i\mathsf{L}A$, choose an inner product (X, Y), and set $\mathsf{P}X = A\,(A,A)^{-1}(A,X)$, $\mathsf{Q} = 1 - \mathsf{P}$. Then, exactly,

$$\dot{A}(t) = i\Omega\, A(t) - \int_0^t \mathrm{d}s\, M(t-s)\, A(s) + F(t), \qquad F(t) = \mathrm{e}^{i\mathsf{Q}\mathsf{L}t}\,\mathsf{Q}\,i\mathsf{L}A, \tag{15}$$

with $i\Omega = (A, i\mathsf{L}A)(A,A)^{-1}$ and memory $M(t) = (F, F(t))(A,A)^{-1}$ (Kubo's second FDT [27]). The random force satisfies $(A, F(t)) = 0$, so the correlation $C(t) = (A, A(t))$ obeys

$$\hat{C}(z) = \frac{1}{z - i\Omega + \hat{M}(z)}\, C(0), \tag{16}$$

which again has Dyson form.

4.2 Worked example: the resonant level

Take $H = \varepsilon c^\dagger c + \sum_k \varepsilon_k d_k^\dagger d_k + \sum_k (V_k c^\dagger d_k + \text{h.c.})$, $A = c$, and the anticommutator inner product $(X,Y) = \langle\{X^\dagger, Y\}\rangle$, so that $(c,c) = 1$. Then $C(t) = \langle\{c^\dagger, c(t)\}\rangle$ and $G^R(t) = -i\theta(t)C(t)$. One finds $i\Omega = -i\varepsilon$ and $F(0) = -i\sum_k V_k d_k$. The Q-projected dynamics evolves d_k freely, so $M(t) = \sum_k |V_k|^2 \mathrm{e}^{-i\varepsilon_k t}$. Continuing $z \to -i\omega + 0^+$ in Eq. (16),

$$G^R(\omega) = \frac{1}{\omega - \varepsilon - \Sigma^R(\omega)}, \qquad \boxed{\Sigma^R(\omega) = -i\,\hat{M}(-i\omega + 0^+) = \sum_k \frac{|V_k|^2}{\omega - \varepsilon_k + i0^+}.} \tag{17}$$

This is the exact hybridization self-energy. For interacting systems the same construction reproduces the Dyson equation: the static (Hartree–Fock) part of Σ sits in Ω and the dynamical part in M [28]. Mori's continued fraction, obtained by iterating the projection, is the recursion method for G^R.

4.3 What Mori does *not* give you, and where Keldysh comes in

Equation (16) determines response functions, that is, the retarded sector. In equilibrium, the FDT then fixes the Keldysh component, $G^K = (1-2f)(G^R - G^A)$. Out of equilibrium, occupations, currents and noise live in $G^<$ (equivalently G^K), which the retarded function alone does not determine. In steady state without bound states, $G^< = G^R \Sigma^< G^A$ [29, 30]. Three routes supply the missing information:

1. *Density-matrix projection* (NZ, Section 3), which evolves the one-time state.
2. *Kadanoff–Baym equations* [23, 31] for two-time $G^{\lessgtr}$, reduced to one-time kinetic equations by the generalized Kadanoff–Baym ansatz (GKBA) [32]:

$$G^{\lessgtr}(t,t') \approx i\big[G^R(t,t')\, G^{\lessgtr}(t',t') - G^{\lessgtr}(t,t)\, G^A(t,t')\big]. \tag{18}$$

   GKBA plays the role of the projection onto one-time quantities, here the one-body density matrix $\rho^{(1)}(t) = -iG^<(t,t)$.
3. *Time-dependent Mori projections* with nonstationary reference states [33]. These produce two-time memory kernels and are closer in spirit to Kadanoff–Baym.

4.4 Generalized Langevin equation ↔ Keldysh saddle point

For a linear bath coupled to a coordinate x (Caldeira–Leggett / Ford–Kac–Mazur [34, 35]), apply a Hubbard–Stratonovich decoupling to the qq noise term in Eq. (8) with a real field ξ. The equation of motion $\delta/\delta x^{\mathrm{q}} = 0$ is then the quantum Langevin equation

$$m\ddot{x}^{\mathrm{c}} + V'(x^{\mathrm{c}}) - \int_{-\infty}^{t} \mathrm{d}s\, \chi(t-s)\, x^{\mathrm{c}}(s) = \xi(t), \qquad \langle \xi(t)\xi(s)\rangle = S(t-s), \tag{19}$$

which is Mori's Eq. (15) for $A = x$, with $M \leftrightarrow \chi$ and $F \leftrightarrow \xi$. Kubo's second FDT and the equilibrium relation between Σ^K and Σ^R are the same statement [1, 7].

5 Dictionary

Keldysh / NEGF	**Nakajima–Zwanzig / TCL**	**Mori**
Contour branches $\pm$; c/q rotation	Left/right superoperators; $A^{\mathrm{c}} = \frac{1}{2}\{A, \cdot\}$, $A^{\mathrm{q}} = [A, \cdot]$	—
D^R (bath response)	χ on q–c lines	damping kernel in M
D^K (bath noise)	S on q–q lines	random-force statistics $(F, F(t))$
Dyson: $G = G_0 + G_0 \Sigma G$	$\hat{\Pi} = \hat{\Pi}_0 + \hat{\Pi}_0 \hat{K} \hat{\Pi}$	$\hat{C} = [z - i\Omega + \hat{M}]^{-1} C(0)$
Irreducible self-energy	NZ kernel K (irreducible diagrams)	memory function M ($\propto \Sigma^R$ for $A = c$)
Single-particle space	Liouville space of ρ_S (dimension d^2)	span of chosen operators
Retarded *and* Keldysh sectors	full one-time state	retarded sector only (out of eq.)
Kadanoff–Baym $\to$ GKBA	NZ $\to$ TCL (one-time closure)	time-dependent projections
Matsubara leg (initial correlations)	inhomogeneous term	initial-value dependence of F
Saddle point + Hubbard–Stratonovich $\Rightarrow$ Langevin	—	GLE (15)
Local Σ^R, constant Σ^K	GKSL / Lindblad	Markovian $M(t) \propto \delta(t)$

Each approach has its natural territory. Keldysh field theory suits many-body problems, renormalization group, large-N and loop expansions, and self-consistent schemes (e.g. open-system DMFT [36]). NZ/TCL suits small systems, where questions about complete positivity, divisibility and memory are posed directly on Π. Mori suits the choice of relevant slow variables and continued-fraction approximations. Exact numerical bridges exist: memory kernels can be extracted from projection-free inputs [37] or from exact impurity solvers [38, 39], and hierarchical equations of motion (HEOM) [40] and process tensors [41] give independent exact references.

6 A sensing-flavoured example: pure dephasing done three ways

Take a qubit sensor $H_S = \frac{\omega_0}{2}\sigma_z$ with $H_I = \frac{1}{2}\sigma_z \otimes B$, a Gaussian bath, and a factorized initial state. On the coherence $X = |0\rangle\langle 1|$, $S^{\mathrm{q}}X = X$ and $S^{\mathrm{c}}X = 0$ (for $S = \sigma_z/2$). By Eq. (11), every response line then vanishes and only noise lines survive. Summing all pairings gives the exact result

$$\rho_{01}(t) = \rho_{01}(0)\, \mathrm{e}^{-i\omega_0 t}\, \mathrm{e}^{-\Gamma(t)}, \qquad \Gamma(t) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \frac{S(\omega)}{\omega^2} F(\omega, t), \tag{20}$$

with $F = 1 - \cos\omega t$ for free induction decay. For a pulse sequence with sign function $y(s) = \pm 1$, $F(\omega, t) = \frac{\omega^2}{2}\left|\int_0^t y(s)\mathrm{e}^{i\omega s}\mathrm{d}s\right|^2$, the filter function of dynamical decoupling and noise spectroscopy [42, 43]. (Normalizations differ between papers.) The three descriptions compare as follows:

- **Keldysh**: Eq. (20) is exp of the qq part of the influence functional (8). Only D^K enters, so for pure dephasing a sensor measures the symmetrized noise and is blind to χ.
- **TCL**: $\mathcal{G}_{01}(t) = -i\omega_0 - \dot{\Gamma}(t)$ with $\dot{\Gamma}(t) = \int_0^t S(\tau)\mathrm{d}\tau$. TCL2 is *exact* here.
- **NZ**: $\hat{K}_{01}(z) = z - 1/\hat{\tilde{\rho}}_{01}(z)$ in the rotating frame is an *infinite* series in the coupling. This is the cleanest demonstration that "order n" means different things in different representations.

For metrology this matters directly: the time dependence of $\Gamma(t)$ at short times ($\propto t^2$ versus $\propto t$) changes the achievable precision scaling under non-Markovian noise [44].

7 Self-study plan (about 8 weeks, 6–8 h/week)

Each stage has a reading core and a derivation to do on paper. The stages are ordered so that every bridge is crossed only after both of its ends are familiar.

Week	Read	Goal / do
1	Breuer–Petruccione Ch. 9 [10]; de Vega–Alonso (NZ/TCL sections) [45]; Nakajima, Zwanzig originals [8, 9]	Derive Eqs. (9) and (10). **Ex. 1.**
2	Zwanzig Ch. 8 [25]; Mori [24]; Kubo [27]; Forster (memory functions) [26]	Derive the GLE (15); second FDT. **Ex. 2.**
3–4	Kamenev (contour, Keldysh rotation, dissipative oscillator) [1]; Kamenev–Levchenko [2]; Rammer–Smith [30]; Stefanucci–van Leeuwen (contour, Langreth rules) [23]	Keldysh rotation; causality structure; Langreth rules; $G^< = G^R\Sigma^< G^A$. **Ex. 3.**
5	Feynman–Vernon [5]; Grabert–Schramm–Ingold [6]; Weiss [7]; Sieberer–Buchhold–Diehl (Lindblad ↔ Keldysh) [16]	Derive Eq. (8) and the Keldysh action of a Lindblad equation. **Ex. 4.**
6	Schoeller–Schön [11]; König *et al.* [12]; Schoeller review [13]; Leijnse–Wegewijs [14]; Timm [15]	The central bridge: prove $K = \sum$(irreducible diagrams). **Ex. 5.**
7	Timm (TCL to all orders) [21]; Reimer *et al.* [20]; Nestmann–Bruch–Wegewijs [19]; Lipavský *et al.* (GKBA) [32]; Shi–Geva [37]; Cohen–Rabani [38]	NZ ↔ TCL map (14); GKBA as one-time closure. **Ex. 6.**
8	Degen–Reinhard–Cappellaro [43]; Cywiński *et al.* [42]; Chin–Huelga–Plenio [44]	Capstone (Exercise 7); write up a one-page summary of your own.

Optional extensions. Gurvitz–Prager infinite-bias Markovianity [46]; Kidon–Wilner–Rabani exact TCL generator [39]; HEOM [40]; process tensors [41]; open-system DMFT [36]; Grabert's time-dependent projections [33].

Exercises

Exercise 1. *Starting from Eqs. (4) and (9), derive Eq. (11). Show that* $\mathrm{Tr}_S\, K^{(2)}(\tau)X = 0$ *for all* X *and explain this as "the last vertex is always* S^{q}*".*

Exercise 2. *Reproduce* $\Sigma^R = -i\hat{M}(-i\omega + 0^+)$ *for the resonant level. Then add an on-site Hubbard* U *for a spinful level and identify the Hartree term in* Ω*.*

Exercise 3. *For Caldeira–Leggett, derive the quantum Langevin equation both from Mori's projection onto* (x, p) *and from the Keldysh saddle point with Hubbard–Stratonovich noise. Match* $M \leftrightarrow \chi$ *and* $(F, F(t)) \leftrightarrow S$*.*

Exercise 4. *From the GKSL equation, write the Keldysh action and read off* Σ^R *and* Σ^K *for a damped cavity mode. Check that they follow from Eq. (11) in the Born–Markov–secular limit.*

Exercise 5. *Prove that* $\mathcal{P}\mathcal{L}_I(\mathcal{Q}\,\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q})\mathcal{L}_I\mathcal{P}$ *generates exactly the irreducible real-time diagrams. (Hint: expand* $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}$ *in* $\mathcal{L}_I$ *and show that every insertion of* $\mathcal{P}$ *corresponds to a cut with no crossing line.) Verify at fourth order by drawing all diagrams.*

Exercise 6. *For the pure-dephasing model of Section 6, compute* $\hat{K}_{01}(z)$ *to fourth order and check Eq. (14). Then do the damped Jaynes–Cummings model and locate the singularity of* $\mathcal{G}(t)$*.*

Exercise 7 (Capstone)**.** *Resonant level between two wide-band leads* (Γ_L, Γ_R)*, prepared empty at* $t = 0$*. (a) Keldysh: compute* $n(t) = -iG^<(t, t)$ *exactly. (b) NZ: extract the exact memory kernel for* $n(t)$ *and show that it becomes time-local in the infinite-bias limit [46]. (c) TCL: build* $\mathcal{G}(t)$ *from Eq. (14) and compare with its perturbative truncations. (d) Sensing: replace the leads by a Gaussian spin bath coupled via* σ_z *and relate your kernel to the filter-function formula (20) under a Hahn echo.*

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